# Ipek Naz Obek

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## Education

**B.A., Cognitive Science** — University of Pennsylvania, Philadelphia, PA — May 2025
Concentration: Computation and Cognition
Minors: Data Science, Computational Neuroscience
*Summa Cum Laude; Distinction in Cognitive Science*
Benjamin Franklin Scholar; Certificate in French
**Honors Thesis:** *Inferring Level Progress in Goal-Directed and Open-Ended Tasks*
**Advisors:** Dr. Alison Gopnik, Dr. Sharon Thompson-Schill, Annya Dahmani
**Relevant Coursework:** Programming Languages, Probability, Data Mining, Statistical Inference, Calculus I & II, Linear Algebra, Cognitive & Computational Neuroscience, Judgment & Decisions, Behavioral Economics, Developmental Psychology

## Research Experience

**Computation, Learning, and Development Lab** — Boston University, Boston, MA — Aug 2025–present
*Research Technician and Lab Manager*
PI: Dr. Kate Nussenbaum

***The development of goal commitment*** (in progress; with Eleanor Holton): examines how goal pursuit strategies, and attention to current goals, change from childhood to adulthood.

- Adapted and programmed a goal-pursuit task (Holton et al., 2024) in jsPsych for online testing across development.
- Fit and compared computational models of alternative persistence strategies in Python; found that although children and adults performed similarly, adults persisted more with their current goals, and stronger attention to current goals predicted greater persistence.
- Analyzed trial-level choice data with mixed-effects models in R; presented at Flux 2026, manuscript in preparation.

***Sunk costs and persistence*** (in progress; with Dr. Brian Sweis): a translational study adapting a rodent foraging paradigm to ask how time already invested shapes decisions to persist or quit.

- Implemented the Web-Surf Task, in which participants decide whether to wait for videos to load or skip to the next offer under a limited time budget.
- Identified that participants rarely quit, limiting tests of sunk-cost effects; refining the task by introducing unexpected changes in waiting time to elicit more informative quitting decisions.

***Learning progress and exploration*** (in progress; with Dr. Julia Leonard): an independent project testing whether exploration is guided by learning progress, by progress relative to one's expectations, or both.

- Designing an experiment in which participants predict their performance across environments that differ in learnability before choosing where to explore.
- Running model simulations to verify that the design can separate the contributions of learning progress and learning-progress prediction errors to choice.

***Lab management and mentorship***

- Mentor an undergraduate honors thesis investigating whether prior environmental volatility influences decisions to persist or give up.
- As the lab's first lab manager, established experimental, recruitment, and data-collection infrastructure, recruiting over 1,000 participants into the lab's database for developmental studies.
- Organize a biweekly multi-lab journal club bringing together researchers to discuss current papers.

**Gopnik Cognitive Development and Learning Lab** UC Berkeley, Berkeley, CA Jun 2024–May 2025
*Research Assistant*
PI: Dr. Alison Gopnik; Advisor: Annya Dahmani

***Inferring learning progress in humans and machines*** (honors thesis; with Annya Dahmani): asks how people infer how much someone has learned from their choices, and whether AI models make the same inferences.

- Designed a novel behavioral paradigm using Procgen environments (the Leaper game), in which participants saw which level a player started on and chose next, then judged how much of the first level the player had completed.
- Found that people used a learner's goal to interpret ambiguous choices: when a learner repeated a level, observers who knew the goal inferred substantially more progress than those who did not.
- Compared human judgments with reinforcement-learning agents and large language models, which did not consistently capture information about goals.

***Curriculum learning in children***: examines how children choose between familiar activities and new challenges as they construct their own learning curricula.

- Designed and analyzed experiments examining how people construct and evaluate curricula for complex, multi-step tasks.
- Supported by the Pinkel Research Fund, Benjamin Franklin Research Grant, and Career Services Research Grant; received the Benjamin Franklin Talk Prize.
- Recruited families on-site and conducted in-person behavioral testing with children in a museum setting, administering experimental protocols and collecting developmental data.

**Thompson-Schill Lab** University of Pennsylvania, Philadelphia, PA Jan 2024–May 2025
*Research Assistant*
PIs: Dr. Sharon Thompson-Schill, Dr. Anna Jenkins

- Conducted honors thesis research, jointly advised across the Thompson-Schill and Gopnik Labs, on human and AI representations of learning progress in goal-directed and open-ended environments.
- Participated in weekly joint lab meetings, discussing current research and literature with faculty and lab members.

**Center for Neuroengineering and Therapeutics** University of Pennsylvania, Philadelphia, PA Jun–Sep 2023
*Research Assistant*
PI: Dr. Brian Litt; Advisor: Dr. James Gugger

***Lesion characteristics and post-traumatic epilepsy*** (MindCORE summer fellowship): examines how the size and location of traumatic-brain-injury lesions relate to later epilepsy.

- Investigated relationships between traumatic-brain-injury lesion characteristics and subsequent post-traumatic epilepsy using multimodal neuroimaging data.
- Developed Python-based quantitative and machine-learning analyses relating lesion volume and location to clinical outcomes.
- Built an automated pipeline for MRI lesion segmentation and volumetric analysis, reducing reliance on manual annotation workflows.

## Honors & Awards

**Distinction in Cognitive Science** University of Pennsylvania 2025
Awarded for completing an honors thesis, "Inferring Level Progress in Goal-Directed and Open-Ended Tasks," and for academic excellence in the major. Thesis advised by Alison Gopnik and Annya Dahmani (UC Berkeley Psychology) and Sharon Thompson-Schill (Penn Psychology).

**Summa Cum Laude** University of Pennsylvania 2025
Graduated with highest Latin honors from the College of Arts & Sciences.

**Benjamin Franklin Talk Prize** — Benjamin Franklin Scholars, University of Pennsylvania — 2025

Recognized for the best senior research presentation among the 2025 cohort of Benjamin Franklin Scholars, Penn's interdisciplinary honors program.

**Pinkel Research Fund** — University of Pennsylvania — 2025

Research funding awarded to support my senior honors thesis on curriculum learning in humans and machines.

**Lucid Communication Challenge, Engineering and Physical Sciences Talk** — Penn National Research Conference — 2024

Selected from a field of applicants at the National Research Conference at Penn to present my honors thesis research as a talk.

**Benjamin Franklin Research Grant** — Benjamin Franklin Scholars, University of Pennsylvania — 2024

Summer research funding awarded by the Benjamin Franklin Scholars Program; supported my research in the Gopnik Cognitive Development and Learning Lab at UC Berkeley.

**Career Services Research Grant** — Penn Career Services — 2024

Competitive summer funding awarded by Penn Career Services; supported my research in the Gopnik Cognitive Development and Learning Lab at UC Berkeley.

**MindCORE Fellowship** — MindCORE, University of Pennsylvania — 2023

Selected for a fully funded 10-week summer research program in cognitive science and neuroscience at Penn's hub for the integrative study of the mind.

**Undergraduate Research Mentoring Award** — Penn Undergraduate Research Mentoring Program — 2023

Fellowship awarded through Penn's Undergraduate Research Mentoring Program, which funds first- and second-year students to conduct summer research with a faculty mentor.

**Dean's List** — University of Pennsylvania — 2021–2025

Awarded in every academic year for a GPA of 3.7 or higher.

**Benjamin Franklin Scholar** — University of Pennsylvania — 2021–2025

Member of Penn's interdisciplinary honors program in the liberal arts and sciences.

## Manuscripts in Preparation

- **Obek, I. N.**, Fu, J., Holton, E., & Nussenbaum, K. (in preparation). *Examining the development of goal commitment*.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (in preparation). *Level learning progress*.

## Unpublished Manuscripts

- **Obek, I. N.** (2025). *Inferring level progress in goal-directed and open-ended tasks*. Honors Thesis, University of Pennsylvania. Advised by A. Gopnik, S. Thompson-Schill, and A. Dahmani.
- **Obek, I. N.**, & Flombaum, J. (2021). *Are empathy and concern psychologically distinct, and is this distinction intuitive?* Research paper, Pioneer Academics Research Program.

## Presentations

- **Obek, I. N.**, Fu, J., Holton, E., & Nussenbaum, K. (2026, September). *Examining the development of goal commitment* [Poster]. Flux Society for Developmental Cognitive Neuroscience, La Jolla, CA.

- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2025, October). *Automated curriculum learning in children and machines* [Poster]. Curiosity, Information Seeking, and Exploration Conference, Brown University, Providence, RI.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2025, October). *Automated curriculum learning in children and machines* [Talk]. Big Team Science Conference.
- **Obek, I. N.** (2025, April). *Machines that think like kids* [Talk]. Benjamin Franklin Scholars, University of Pennsylvania, Philadelphia, PA. Awarded the Benjamin Franklin Talk Prize.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2024, September). *Automated curriculum learning in children and machines* [Talk]. Lucid Communication Challenge, Engineering and Physical Sciences session, Penn National Research Conference, Philadelphia, PA.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2024, Fall). *Automated curriculum learning in children and machines* [Poster]. Center for Undergraduate Research and Fellowships (CURF) Poster Symposium, University of Pennsylvania, Philadelphia, PA.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2024, September). *Automated curriculum learning in children and machines* [Poster]. Penn National Research Conference, Philadelphia, PA.
- **Obek, I. N.**, Gugger, J., Narang, S., Vasireddy, M., & Zhao, C. (2023, Fall). *Understanding the link between TBI-induced lesions and epilepsy* [Poster]. Center for Undergraduate Research and Fellowships (CURF) Poster Symposium, University of Pennsylvania, Philadelphia, PA.
- **Obek, I. N.**, Gugger, J., Narang, S., Vasireddy, M., & Zhao, C. (2023, August). *Understanding the link between TBI-induced lesions and epilepsy* [Poster]. MindCORE Research Fellowship Conference, Philadelphia, PA.

## Teaching

**Neuroscience and Society** — Teaching Assistant, Penn Undergraduate Program in Neuroscience — 2024–2025
Instructor: Dr. Sharon Thompson-Schill

Supported students through weekly office hours, in-class instruction, and group activities, and provided individualized academic guidance.

**Introduction to Cognitive Science** — Teaching Assistant, Penn Undergraduate Program in Cognitive Science — Fall 2023
Instructor: Dr. Russell Richie

Led weekly recitations and office hours for 20+ students, developed instructional materials, and guided students through course concepts and assignments.

**MATH 114: Calculus II** — Tutor, Penn Weingarten Tutoring Center — Fall 2022

Tutored first-year students for 4 hours per week, providing individualized support with mathematical problem-solving and study strategies.

## Mentorship

**Benjamin Franklin Scholars Mentor Program** — Peer Mentor, University of Pennsylvania — 2024–2025

Mentored 7 Benjamin Franklin Scholars in their academic and social transition to Penn through regular individual outreach and community-building events.

**Center for Undergraduate Research and Fellowships** — Research Peer Advisor, University of Pennsylvania — 2024–2025

Advised undergraduates pursuing research opportunities and organized research programming, including an international faculty symposium on the experiences and challenges of international academics.

**Pioneer Academics** Program Coordinator and Advisory Board Member 2022–2025

Advised 100+ students through the research and writing process and coordinated research seminars on effective research practices and faculty mentorship.

**Young Guru Academy** Alumni Mentor 2019–present

Mentor Turkish high school and university students pursuing international education and STEM careers. Led STEM outreach across 5+ cities in rural Anatolia, training teachers and students and delivering hands-on science education to underserved communities; presented at YGA's annual summit to ~20,000 attendees.

## Leadership & Service

**Cognitive Science Student Association** Founder & President, University of Pennsylvania 2024–2025

Founded and led Penn's Cognitive Science Student Association, organizing faculty talks, research seminars, mentoring, and networking initiatives connecting students with faculty, alumni, and industry professionals.

**The DREAM Program** Recruitment Chair, Philadelphia, PA 2023–2025

Led volunteer recruitment and weekly educational programming for preschool-aged children in West Philadelphia, including hands-on science activities designed to foster curiosity, collaboration, and confidence in learning.

**Turkish Student Association** President, University of Pennsylvania 2022–2025

Directed a 50-member organization and community initiatives, including raising $8K for earthquake-relief efforts.

**Assembly of International Students** Social Committee Member, University of Pennsylvania 2021–2025

Organized social and cultural programming to foster community among Penn's international student population.

**The Daily Pennsylvanian** Photographer, University of Pennsylvania 2021–2025

Photographed campus events and contributed visual content for multimedia stories at Penn's student newspaper.

## Skills

**Programming:** Python, R, JavaScript, MATLAB, SQL, Java, OCaml

**Research tools & platforms:** jsPsych, PsychoPy, Qualtrics, Prolific, Pavlovia, Git/GitHub, Jupyter, LaTeX, AFNI

**Certifications:** CITI Training (Biomedical Responsible Conduct of Research; Human Subjects Research; Social-Behavioral Research with Children)

**Languages:** Turkish (native), English (fluent), French (fluent)

## References

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**Dr. Sharon Thompson-Schill**
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University of Pennsylvania
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**Dr. Russell Richie**
Director of Strategy and Impact
Progress and Poverty Institute
Formerly Associate Director, Cognitive Science and MindCORE, University of Pennsylvania
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